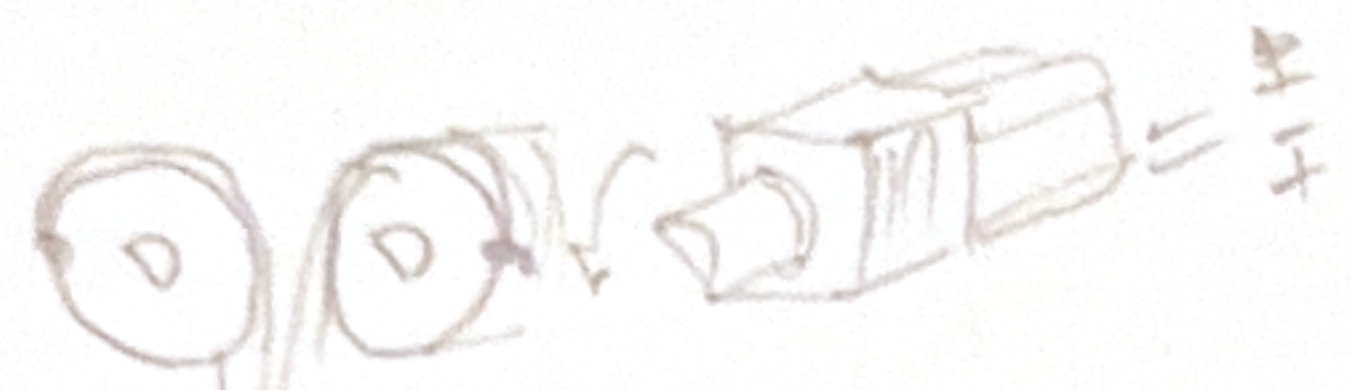
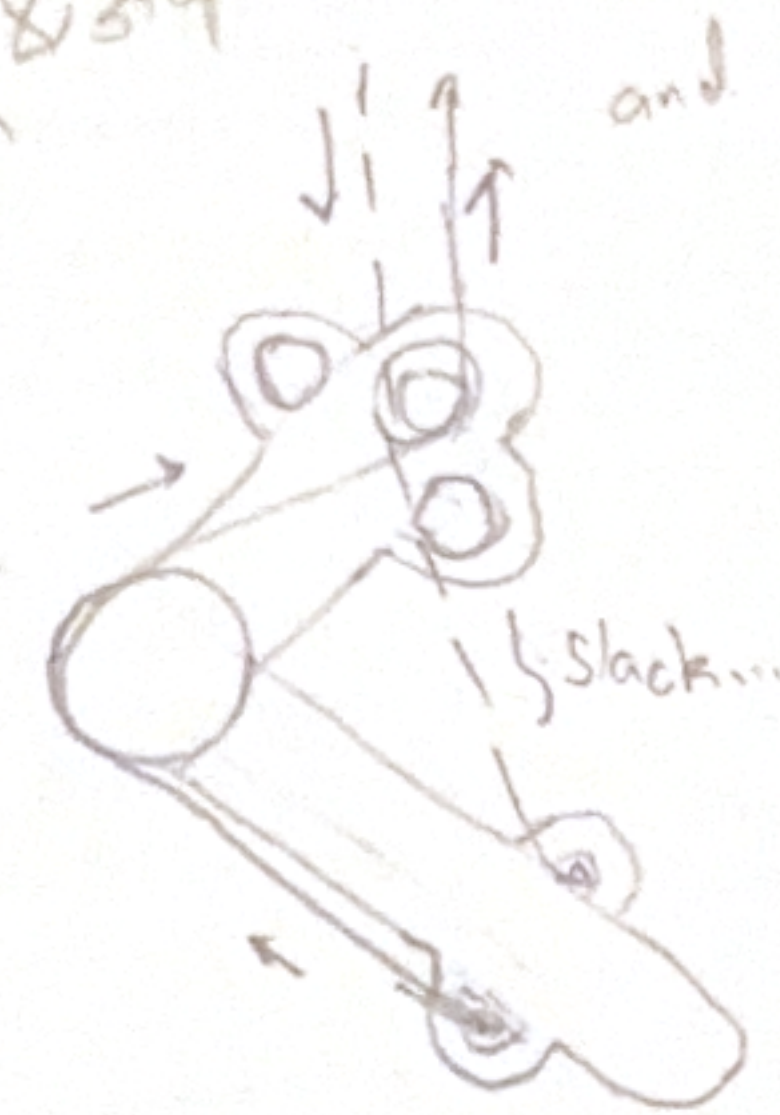


19/29



mount & stop tendon



and vice versa

→ ought to be kept in slight tension

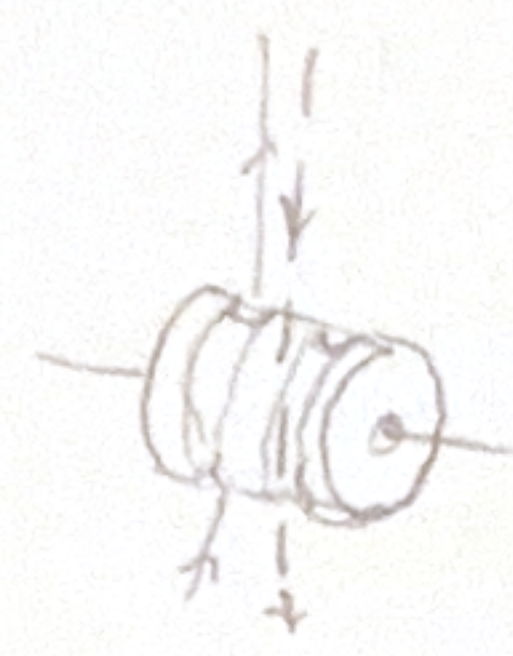
Actually.



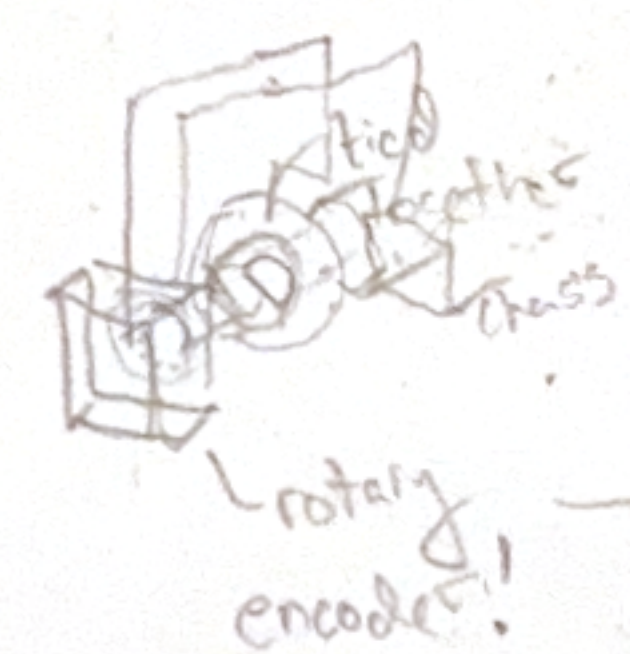
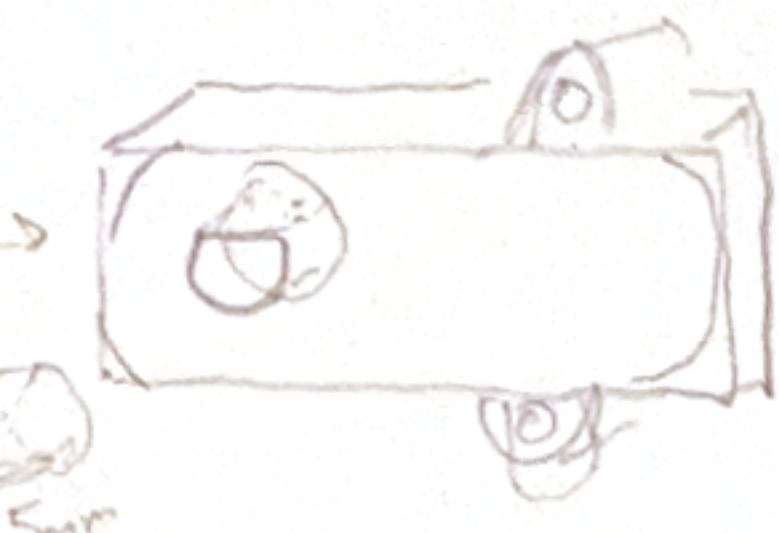
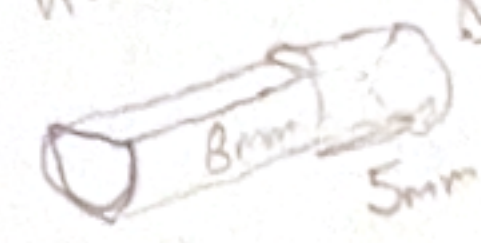
two smooth pins (instead of a pulley/loop)

> Might tangle

> No just print actual pulley, pal

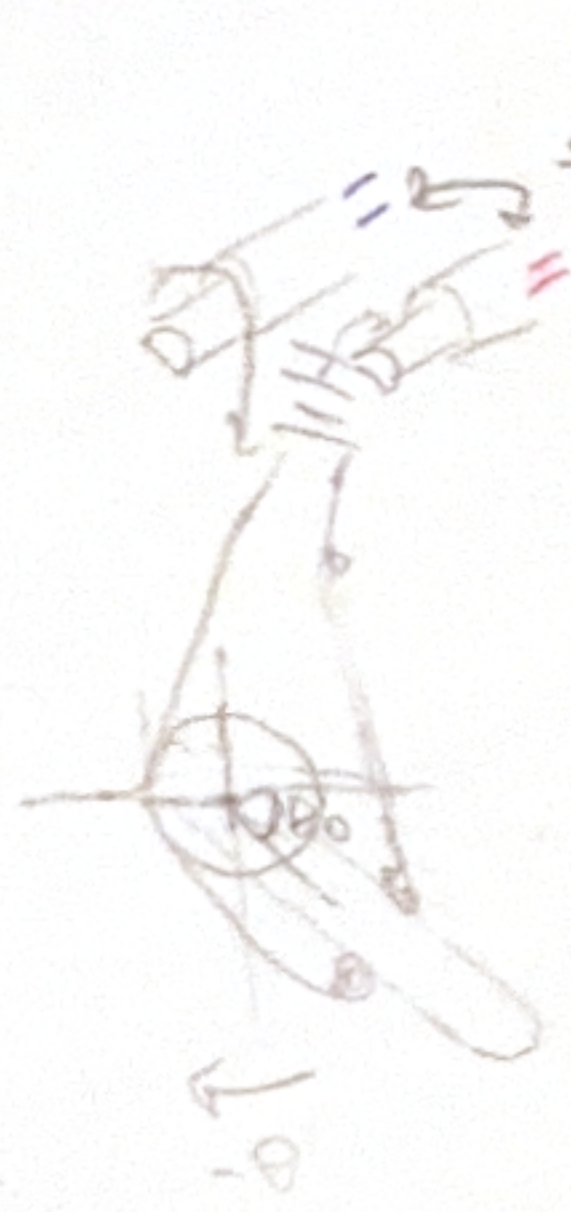
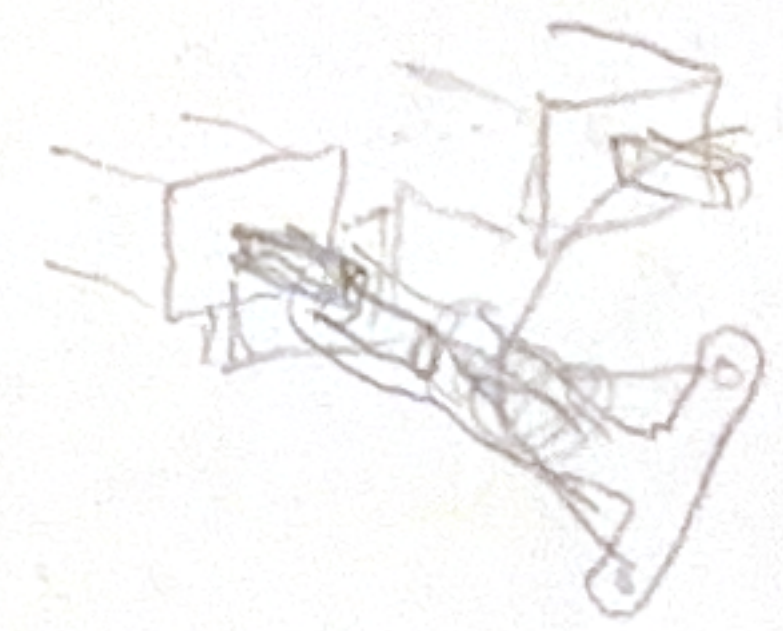


hole



can implement feedback in situ with μC code

- Options:
- 1) Control circuitry, blind motors
 - 2) Make motors "smart" to μC

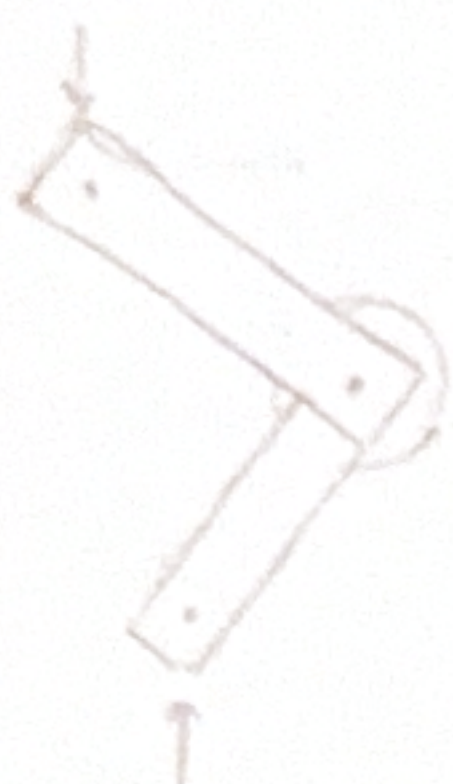


steady state
"fighting"
Pot



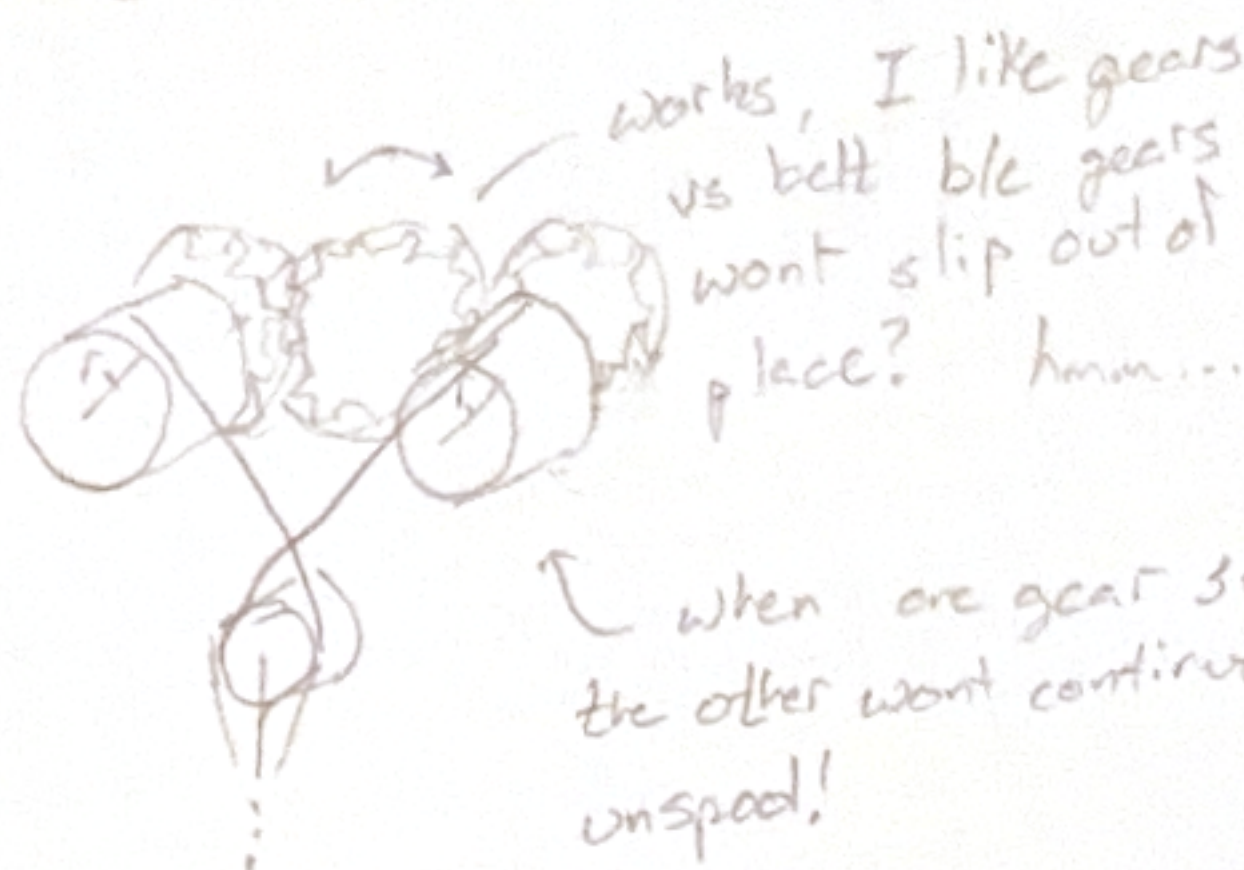
I bet I could make a feedback circuit that would let me cheat with "blind motors"

(a/30)



moment about K?

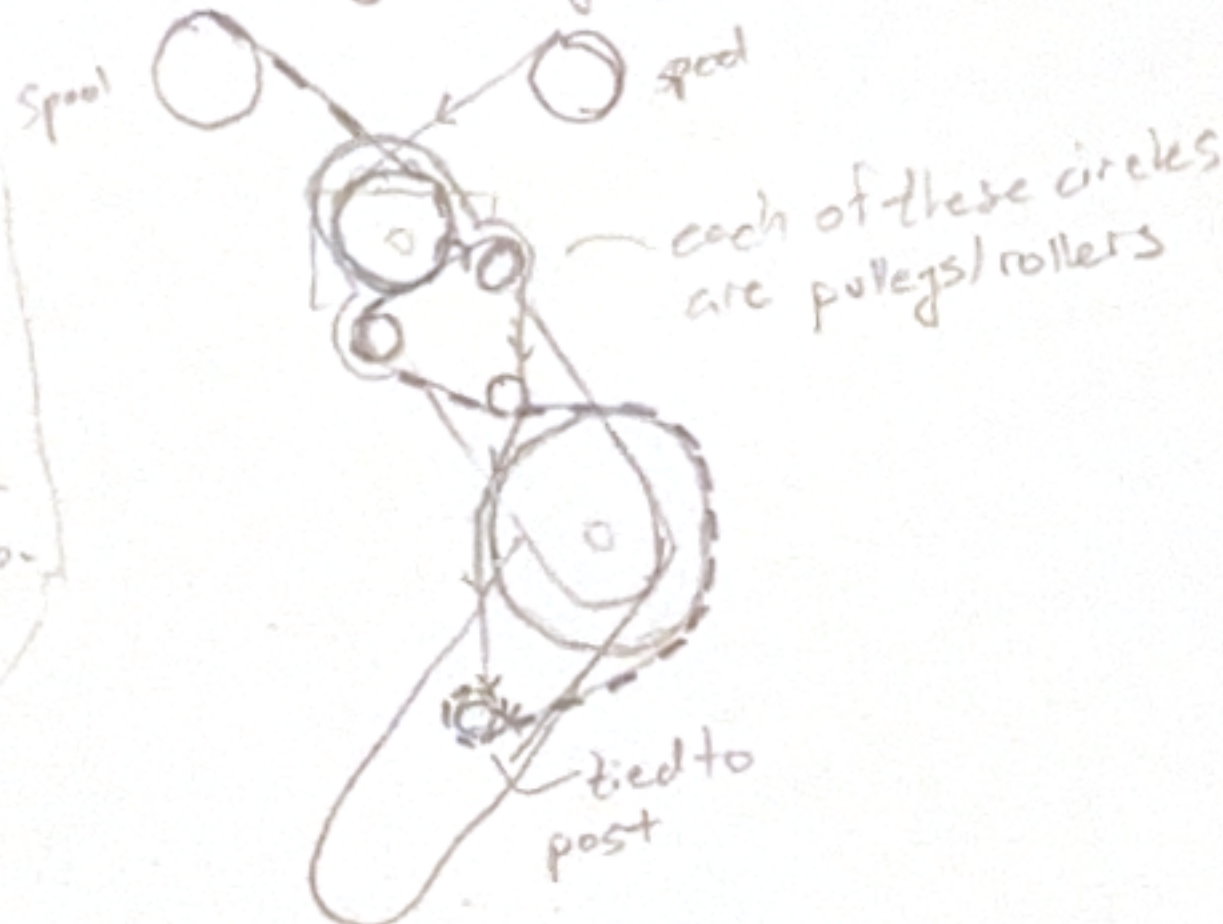
proto3(red)
Didn't think,
don't need independent
motors for tendons



works, I like gears
vs belt b/c gears
won't slip out of
place? hmmm...

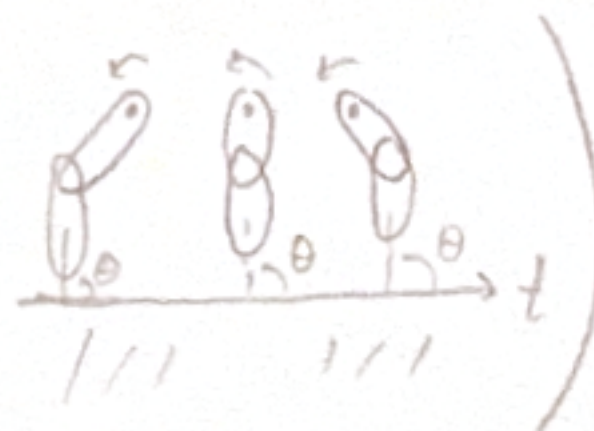
When are gear sizes,
the other won't continue to
unspool!

Big routing / pulley plus ✓



String hardly slips now

(note: Interesting side effect
of the tension means, when held
in place, the hip rotation keeps
the knee angle relative to ground
the same.

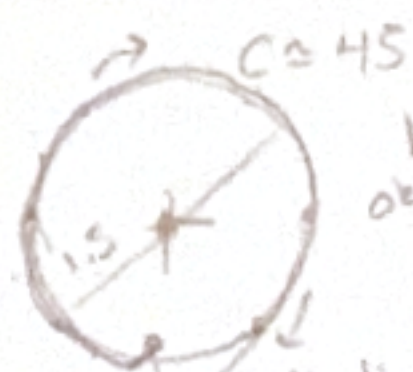
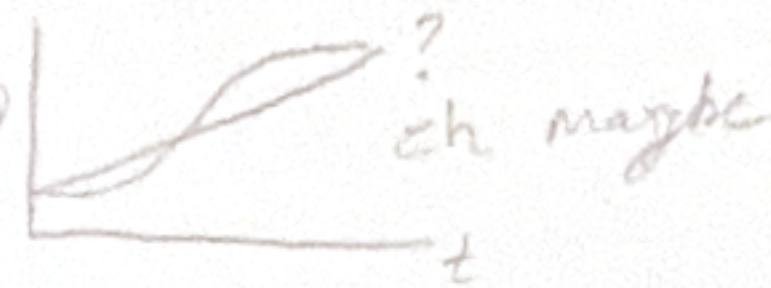


2 spools vs 1

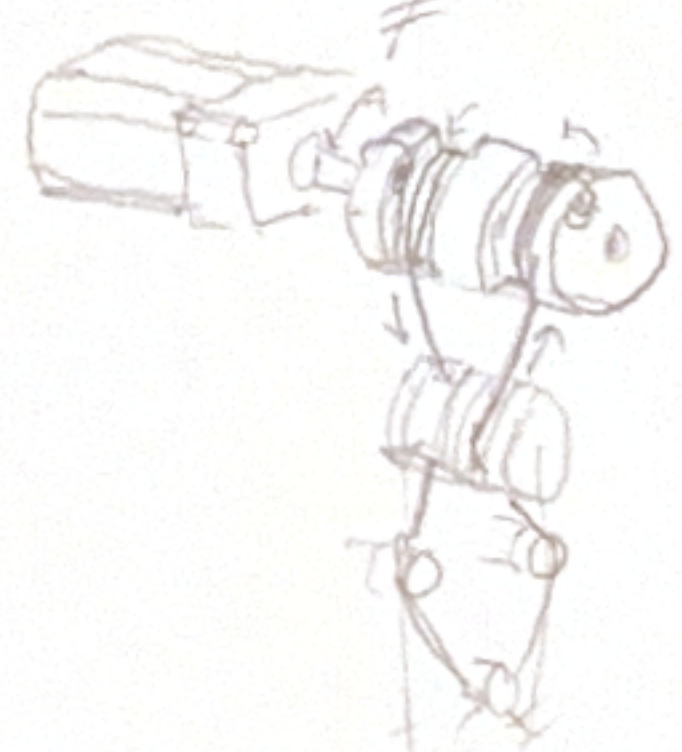


$\frac{1}{2}C = 45\text{mm}$! big!
 $90\text{mm} = \pi d \rightarrow d \approx 30\text{mm}$

would it be linear motion even?

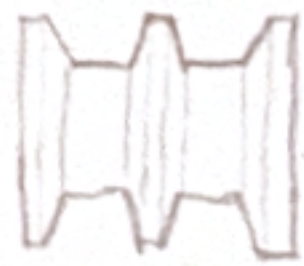


hmm
de, I don't hate that



heck size
could be arbitrary
if we use a
serial-spool

hmm
could be a pain to
tension?

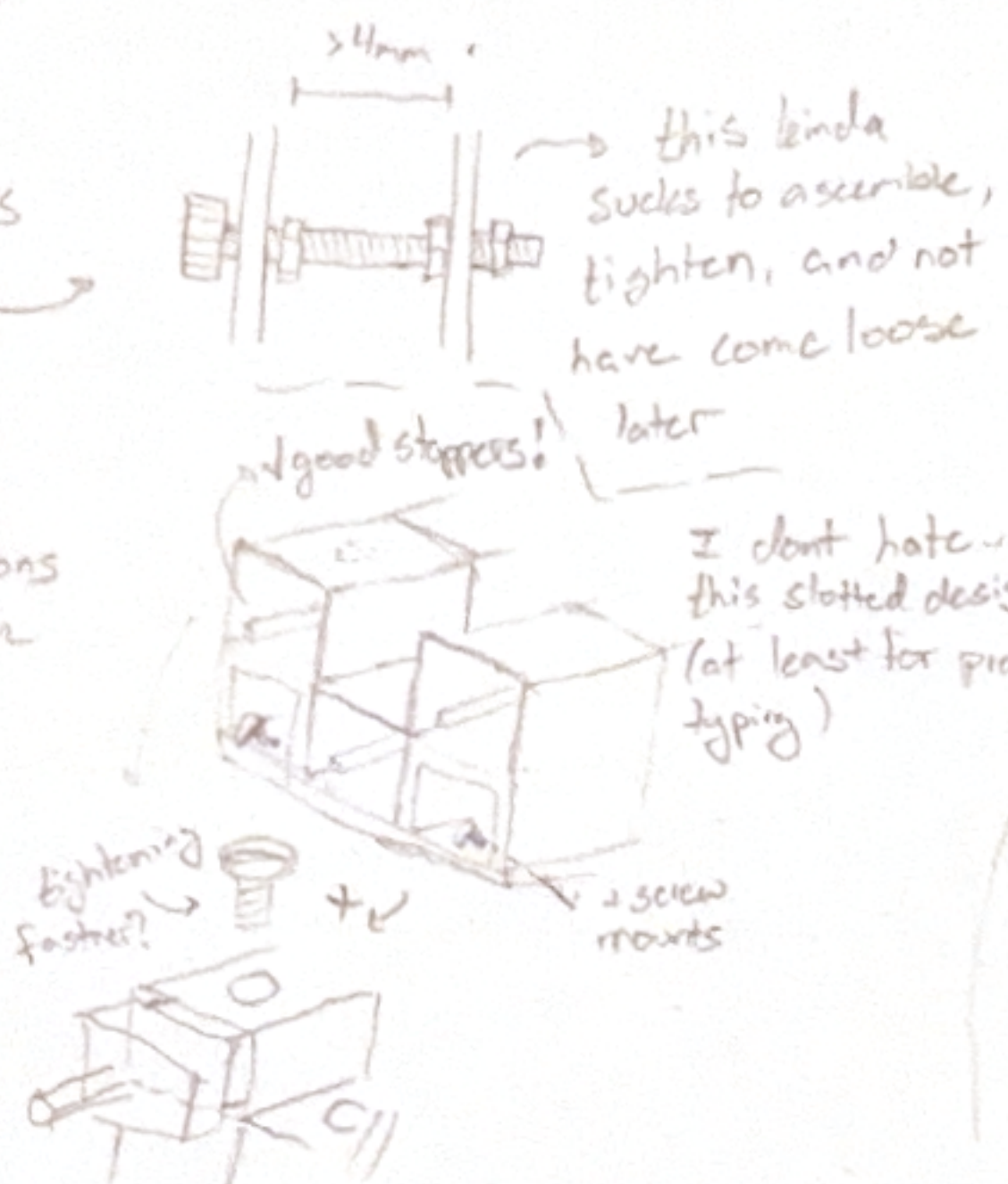


Issues w/ proto 2 (black)

- Line keeps slipping off pulley
- Either needs constant tension
or a better routing path / design
or both

Awkward to screw together legs
with multiple nuts

Awkward motor housing -
ought to consider motor dimensions
a bit before trying to bolt them
on ~



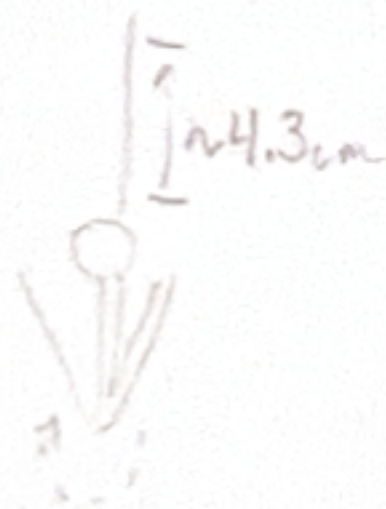
this kinda
sucks to assemble,
tighten, and not
have come loose
later

I don't hate
this slotted design
(at least for proto-
typing)



tightening
fastener?

2 screw
mounts



$$C = 2\pi r = \pi d$$

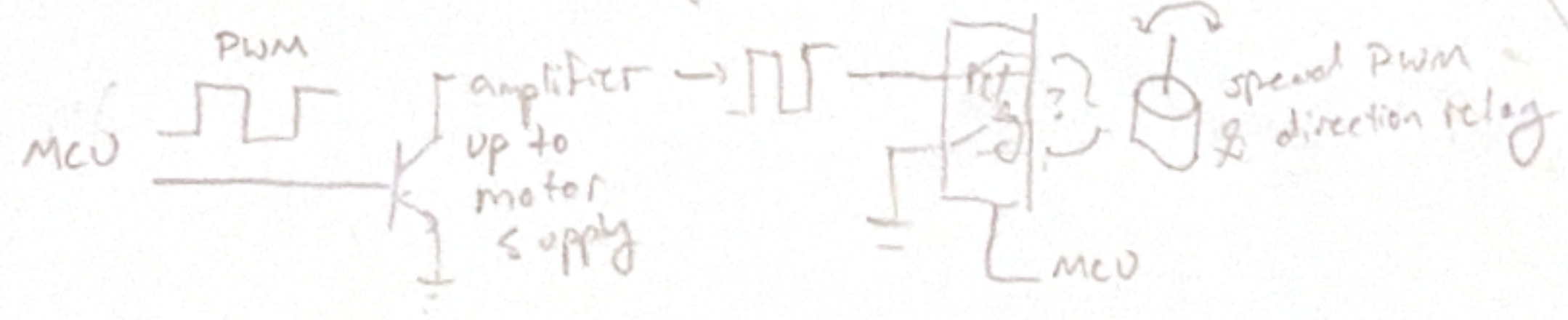
$d = 12\text{mm}$
 $C \approx 37.7\text{mm} < 41.3\text{mm}$
 $C = 45\text{mm} \rightarrow d \approx 14.3\text{mm}$

(10/1)



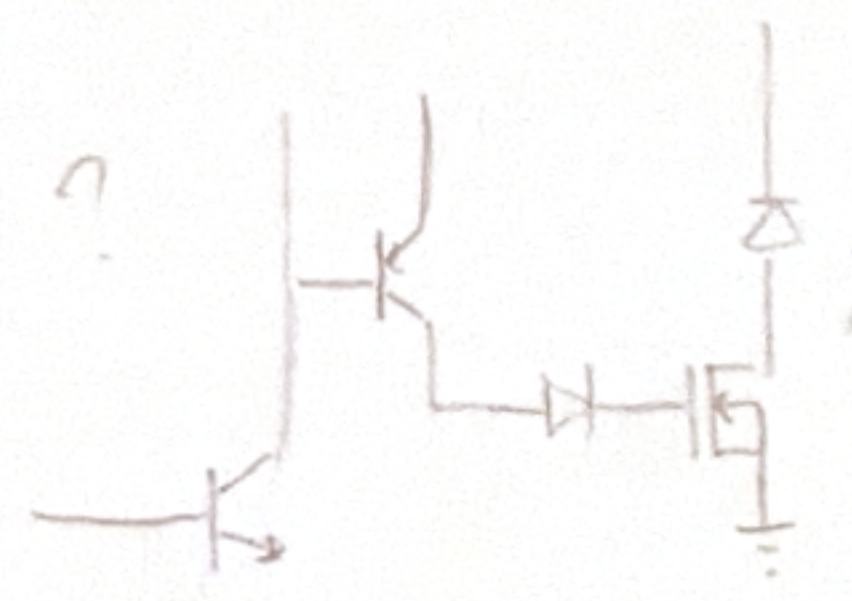
MOTOR SPEED & DIRECTION CONTROL

basically ~

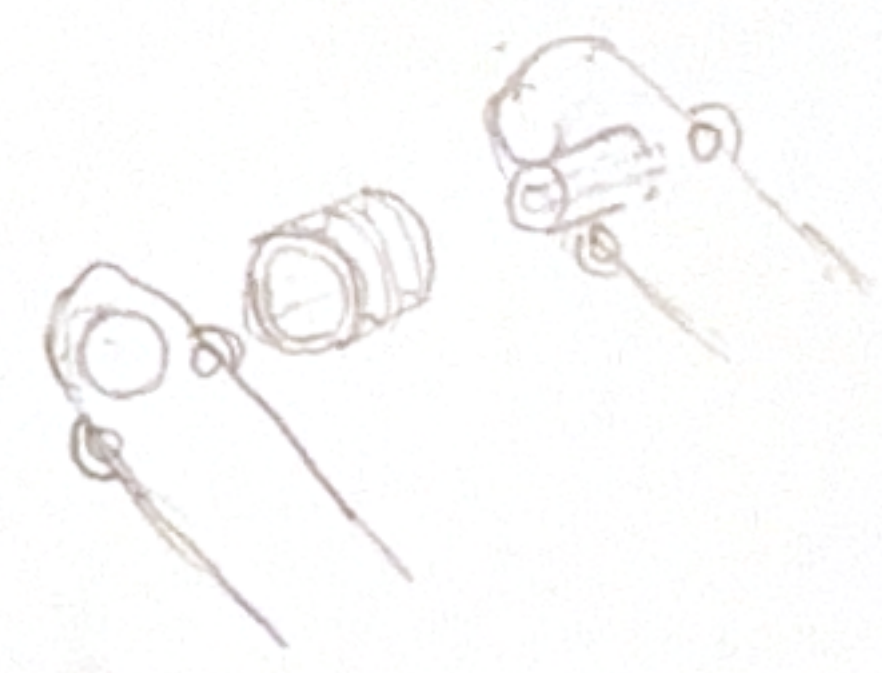
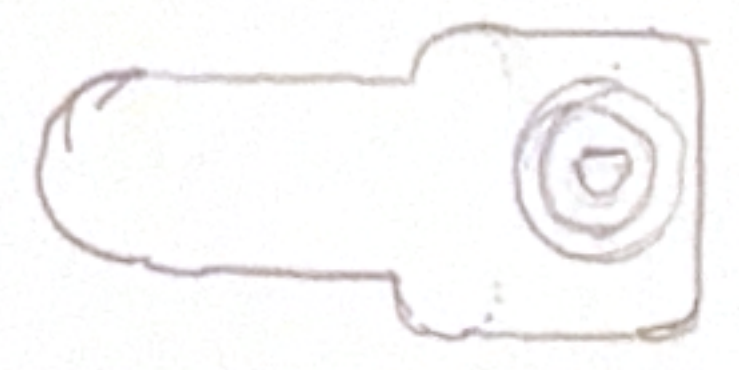
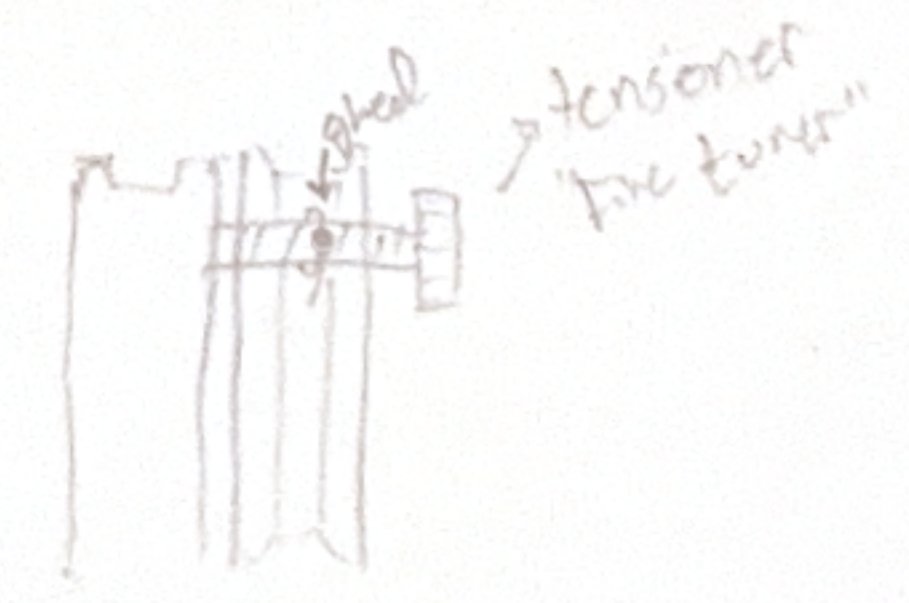


Online CKT actually PWM-ing the motor's ground path
Correction - pretty a good idea tbh?

3.3V logic
eg, 5VDC Motor 3.3V

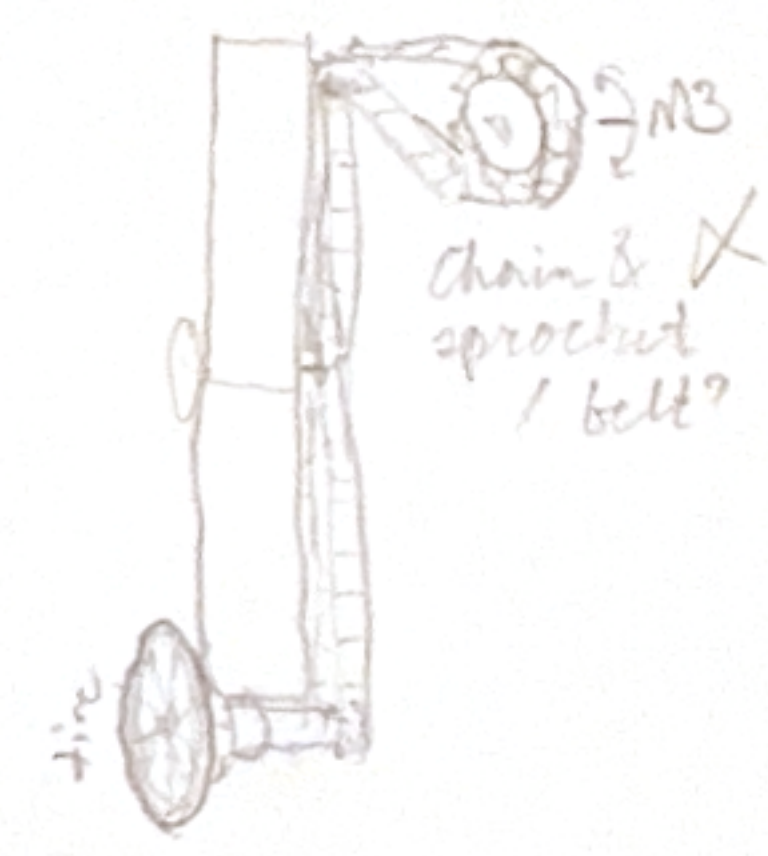


Internet wants
me to use MOSFETS



tensioner & belt/pulley
google:
"lawn mower blade
belt pathing"
-house of mech-e's
• arm/lever w/ spring
to keep tension

~ later?



$$3kg \cdot 9.8m/s^2$$

$$(30cm)(3kg)(9.8m/s^2) + (3kg)(9.8m/s^2)(30cm)$$

$$= 17.64 Nm$$

say M1

$$\left(\frac{24 RPM}{25 kgf \cdot cm} = 7.215 Nm \right)$$

1.08W

ok like, we might be limited
to the 4-5 Nm range (\$\$\$)

Load
59 RPM
1.0 kg/cm

Future Collg:
look up -
"High torque self-locking
worm gear encoding motor"
(on ali express)

if we need like 5x torque,
so 5x diameter $\frac{1}{5} \times RPM$

15cm diameter knee pulley
↳ 3cm Ø spool
↳ 41.8 RPM ... M1